2× MSCI ACWI SMA-gated strategy — best window

The all-world analogue of the SPY-200-SMA / SPXL strategy, at 2× leverage · modeled history 1970-02-02→2026-07-07 · generated from research/sma_sweep_acwi.py

> **Conclusion — use the ~250-day SMA** (robust ridge ≈ 230–270d; notably *longer* than the US 200-day because at 2× leverage whipsaw and volatility-decay punish a fast gate). **What's robust** (survives every fix in the review, all costs/taxes, and the bootstrap): the *window choice* and the **drawdown control** — the gate **more than halves** the 2× drawdown (-46% vs an ungated 2× ACWI's catastrophic -87%) and beats ungated-2× Sharpe in **88%** of Monte-Carlo paths. **What's softer** (single-path, in-sample): the absolute return edge. On the modeled path the gate's pre-tax CAGR is 14.6%; **after 10 bp switch costs + German tax it is 11.4%** — still ahead of ungated 2× (11.3%), though the turnover tax shrinks the pre-tax edge. The US 200-day is too fast here (Sharpe 0.494); your 255-day World number sits inside the ridge (Sharpe 0.545), so reusing 255 is fine. **Read §10 (limitations) before acting** — the headline numbers are modeled, frictionless upper bounds.

Glossary (every metric used below)
CAGR — compound annual growth rate (annualised return). · **Vol** — annualised volatility (std-dev of daily returns × √252); higher = bumpier. · **Sharpe** — (CAGR − 2% risk-free) ÷ Vol; return per unit of risk, higher = better. · **MaxDD** — worst peak-to-trough loss ever suffered (more negative = worse). · **Calmar** — CAGR ÷ |MaxDD|; return per unit of worst-case pain. · **InMkt** — % of days the gate was invested (vs in T-bills). · **Switch** — number of in↔out trades over the whole period (turnover; each is a taxable event). · **median / p5 / p95** — the middle / 5th / 95th percentile across the Monte-Carlo simulations. · **P_best** — share of simulations in which that window had the highest Sharpe. · **SMA** — simple moving average; the gate is "in" when price > SMA×1.01, "out" when < SMA×0.99.

1 · Data sources

Series	Source	Span used	Role here
MSCI ACWI gross index (USD)	MSCI end-of-day API (index 892400)	2001–2008	real ACWI daily returns
iShares MSCI ACWI ETF (ACWI.US)	EODHD	2008→today	real ACWI daily returns (live + future)
MSCI ACWI net TR (EUR), monthly	Curvo (curvo.eu), 1987+	1988–2001	real monthly anchor (embeds true EM weights)
MSCI World (URTHSIM → URTH)	Testfolio sim spliced w/ real URTH (EODHD)	1970–1987	ACWI predecessor segment, used as-is (index didn't exist yet; EM <1% at 1988 inception)
MSCI World (same series)	—	1988–2001	intra-month daily *texture* only

Series	Source	Span used	Role here
Xtrackers S&P 500 2x Lev. Daily Swap (XS2D)	EODHD (LSE, USD)	2010→today	real-product cost calibration of the synthetic 2× sleeve
EUR/USD spot	FRED DEXUSEU (1999+) + Deutsche Mark EXGEUS (pre-1999)	1988→today	convert Curvo EUR→USD
3-month US T-bill rate	FRED TB3MS → DGS3MO	1970→today	LETF financing cost + the "out" (cash) leg

The first two are the genuine ACWI; Curvo is real ACWI too (just EUR/monthly, FX-converted); World is used *only* to give the pre-2001 series daily wiggle — never as the return itself.

2 · How the pre-2001 index is modeled (EM-share-correct)

Real MSCI ACWI daily only starts 2001 — too short for a leveraged backtest. A MSCI-World-only proxy is **wrong** pre-2001: it omits emerging markets, whose ACWI weight evolved a lot over time — so over 1988-2000 World-only understates ACWI by ~0.34%/yr.

year	1988	1997	2002	2010	2021	today
EM weight in ACWI	<1%	6.8%	~4%	13.5%	13%	~10%

Instead we anchor to the **real ACWI monthly history** (which embeds those exact time-varying EM weights by construction). The source is Curvo's **iShares MSCI ACWI UCITS ETF (Acc)** series — a real fund-NAV track record (monthly, EUR), not the bare index — which we FX-convert to USD. **Validation:** converted-Curvo vs real MSCI ACWI USD-gross over the 2001+ overlap has monthly corr **0.990** (R^2 0.98), return drift only **-0.05%/yr**. The near-zero drift is what you'd expect from a fund-TR series — its ~0.20% expense ratio is already inside that measured drift (so if anything the modeled pre-2001 returns are a hair *conservative*, carrying a small fund fee the bare index wouldn't). Using the real EM weighting adds **+851.3%** to 1988-2000 cumulative growth vs a World-only proxy.

Daily granularity (needed for a daily SMA gate) comes from **temporal disaggregation**: inside each pre-2001 month we take MSCI World's daily shape and rescale it multiplicatively so the month compounds *exactly* to the real ACWI monthly return (reconstruction error ~1e-15). World supplies only the intra-month wiggle; every monthly return is the real EM-weighted ACWI. A literal daily World+EM blend is impossible before 2001 (no daily EM exists pre-2003) — and provably immaterial: tested on 2003+, World-texture and a true World+EM blend give the same gated Sharpe to ±0.02.

The 1970–1987 head. MSCI ACWI only exists from Dec-1987, so "the longest possible ACWI backtest" necessarily starts there — unless one accepts the index's predecessor. We do, explicitly: 1970→1987 uses **MSCI World daily returns as-is**. That is not a stretch — at ACWI's 1988 inception EM weighed **<1%** and the MSCI EM index itself only launched 1988, so an "all-country" investor of the 1970s-80s held, in practice, exactly World. The segment is labeled throughout; nothing pre-1988 is scaled or synthesised. **Full chain:** World as-is (1970→1988) → EM-anchored modeled daily (1988→2001) → real MSCI ACWI gross (2001→2008) → real ACWI ETF (2008→today).

3 · How the 2× sleeve (LETF) is modeled

There is no 2× ACWI ETF with long history, so we synthesise one with a **daily-reset** formula — the same construction (and constants) the production engine uses for its leveraged ETFs (e.g. synthetic WLDU/SPXL):

$$r_{2\times}(\text{day}) = 2\cdot r - SW\cdot(L-1)\cdot(\text{T-bill}_{daily} + \text{spread}/252) - (L-1)\cdot\text{expense}/252$$

In plain words: to get 2× exposure with 1× of your own money you effectively **borrow another 1×** and invest the doubled amount. So each day the sleeve **earns twice the index's move** (the `2·r` term), then pays two bills: (1) the **interest on that borrowed half** — charged at the short-term T-bill rate plus a small ~0.4% spread, and nudged up ~10% (the swap multiplier) for the cost of the swap that delivers the leverage; and (2) the **fund's own running expense** (~0.5%/yr). Two consequences matter: the borrow cost **rises and falls with interest rates** (cheap near 0%, painful when T-bills are 5%+), and because the leverage **resets to 2× every single day**, a sideways-but-choppy market slowly **bleeds value** ("volatility decay") even if the index ends flat — which is precisely why a trend gate that sits out the chop is so valuable at 2×.

term	value	meaning
L (leverage)	2×	daily target exposure to ACWI
2·r	—	twice the underlying ACWI daily return
T-bill (financing)	actual 3m rate, time-varying	cost to borrow the extra (L−1)=1× notional — the key driver
SW (swap multiplier)	1.1	broker/swap markup on the financed notional (Testfolio default)
spread	0.4%/yr	financing spread over the T-bill (≈40 bp)
expense	0.5%/yr ×(L−1)	fund expense ratio drag

Why time-varying financing matters: the borrow cost is the dominant drag and it scales with the short rate, which ran 0%→15% across this window. A flat rate would badly mis-state pre-2008 returns. Total annual drag on the 2× sleeve at different rate levels:

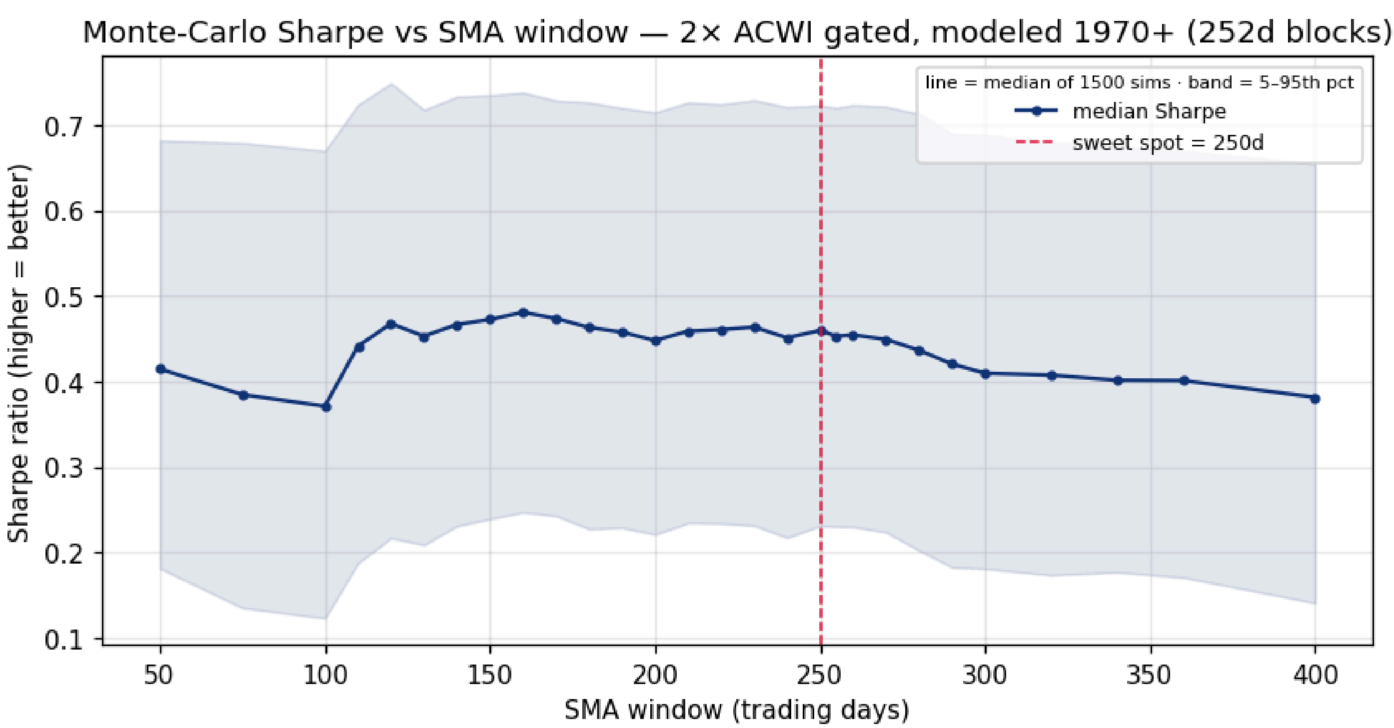
short rate	annual drag on 2× sleeve
T-bill 0%/yr	0.94%/yr
T-bill 4%/yr	5.34%/yr
T-bill 10%/yr	11.94%/yr

Worked example: on a day ACWI returns +1.00% with the T-bill at 4%/yr, the 2× sleeve returns 2×1.00% – 0.019% (financing) – 0.002% (expense) = **1.979%** — slightly under a naïve +2.00%. Daily resetting also causes volatility decay in choppy markets, which is exactly why the SMA gate (which sits out the chop) adds so much value at 2×.

Calibration against a real product. The formula above is no longer purely theoretical: we benchmarked it against the **Xtrackers S&P 500 2x Leveraged Daily Swap UCITS ETF** (XS2D, USD — structurally identical to the upcoming ACWI product: TER 0.60%, unfunded swap) over its full 2010→2026 history. Month-end levels track at correlation 0.95, with the synthetic running **+0.73%/yr richer** than the real fund — i.e. real-world swap costs slightly exceed our textbook assumptions. The estimate is stable, not endpoint-driven: trimming the first 0/6/24 month-ends gives +0.73/+0.74/+0.79%/yr (reproduce via `research/xs2d_calibration.py`). Measured +0.73%/yr, applied as **0.75%/yr** in a **"real-product drag"** scenario that appears as an extra row in the strategy-comparison table (§7) and the after-tax table (§9) — treat those rows as the expectation for the actual ETF. One transfer caveat: the calibration is US-underlying; a 2× ACWI swap index carries a larger dividend-withholding surface, so the real fund could lag marginally more — re-calibrate against its NAV once it trades.

The strategy itself: the SMA gate is computed on the *unleveraged* ACWI index (just as SPXL gates on SPY); when ACWI > its N-day SMA×1.01 we hold the 2× sleeve, when < SMA×0.99 we hold T-bills. Signal checked daily, executed next day, ±1% hysteresis band to avoid flip-flopping.

4 · Window robustness — Monte Carlo (1500 sims, 252-day blocks)



How to read: each window's gate is re-run on 1500 bootstrapped 56-year histories (resampled in ~1-year blocks). The line is the median Sharpe across those sims; the shaded band is the 5th–95th percentile. **The MC is a robustness check, not the selector.**

Block-resampling necessarily shreds part of the multi-month trend structure a long SMA exploits, which tilts the bootstrap argmax toward shorter windows — on this run its argmax is **160d**, versus the deterministic real-path best of **250d**. We select on the real path (trend structure intact), cross-checked by the real-data-only 2001+ sweep and the tax analysis (§9: shorter windows trade ~50% more and lose the difference to the German tax drag). See §10 for the paired test and block-length sensitivity.

Columns: **med/p5/p95_Sharpe** = median & 5–95th-pct Sharpe across sims · **med_CAGR / med_MaxDD** = median return & drawdown · **P_best** = % of sims where this window won. Green row = recommended 250d.

window	med_Sharpe	p5_Sharpe	p95_Sharpe	med_CAGR	med_MaxDD	P_best
50	0.415	0.181	0.682	10.8%	-54.0%	10.7%
75	0.385	0.135	0.679	10.3%	-59.8%	1.3%
100	0.371	0.123	0.670	10.1%	-60.3%	0.1%
110	0.441	0.187	0.724	11.5%	-56.7%	4.7%
120	0.468	0.217	0.749	12.1%	-54.0%	14.1%
130	0.453	0.209	0.718	11.8%	-52.7%	1.5%
140	0.467	0.231	0.734	12.2%	-51.9%	4.4%
150	0.473	0.239	0.735	12.4%	-52.0%	5.3%
160	0.481	0.247	0.738	12.6%	-50.2%	11.6%
170	0.474	0.243	0.729	12.5%	-50.5%	6.7%
180	0.463	0.228	0.727	12.4%	-51.2%	4.0%
190	0.458	0.229	0.720	12.3%	-52.2%	2.3%
200	0.448	0.221	0.715	12.1%	-52.7%	0.2%
210	0.459	0.234	0.727	12.4%	-52.5%	1.9%
220	0.461	0.234	0.725	12.5%	-52.3%	3.4%
230	0.464	0.231	0.729	12.6%	-52.5%	4.4%
240	0.451	0.217	0.721	12.3%	-53.9%	2.5%
250	**0.460**	**0.231**	**0.723**	**12.5%**	**-53.0%**	**5.7%**
255	0.453	0.230	0.720	12.5%	-53.0%	2.3%
260	0.455	0.230	0.724	12.5%	-53.3%	3.2%
270	0.449	0.224	0.722	12.4%	-53.8%	2.3%
280	0.437	0.202	0.713	12.1%	-55.2%	2.1%
290	0.420	0.182	0.690	11.7%	-56.8%	0.7%
300	0.410	0.181	0.688	11.6%	-57.3%	0.2%
320	0.407	0.173	0.680	11.5%	-58.7%	0.7%
340	0.401	0.177	0.675	11.5%	-58.6%	1.3%
360	0.401	0.171	0.670	11.5%	-59.1%	1.6%
400	0.381	0.141	0.655	11.1%	-60.8%	0.9%

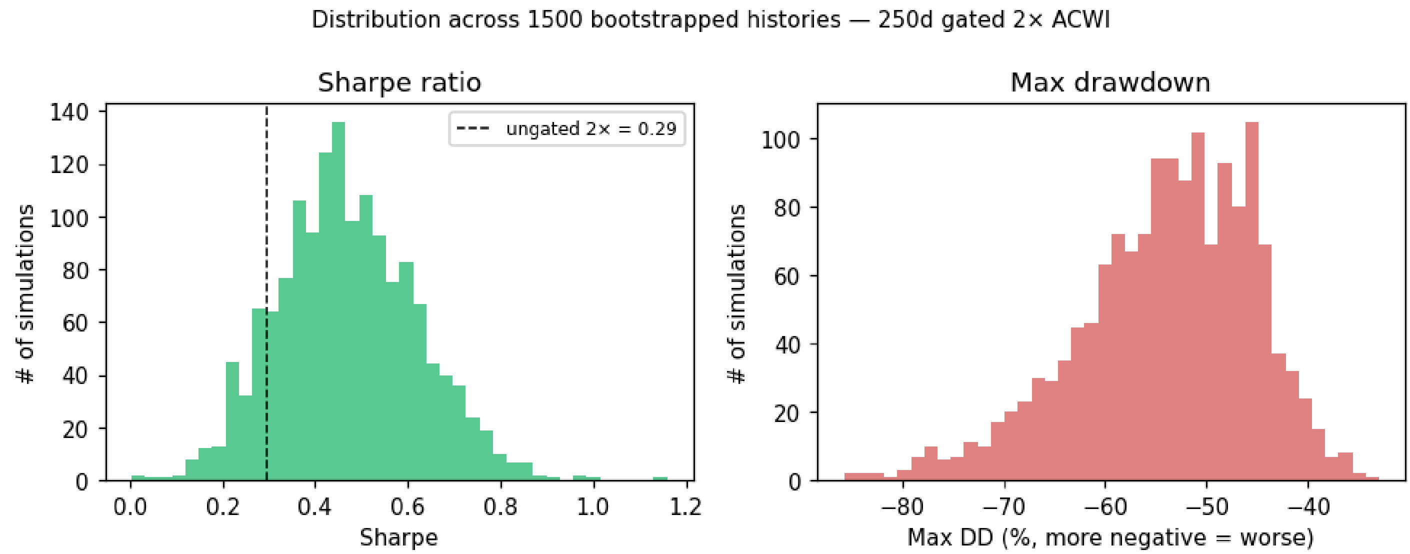
5 · Deterministic sweep — every window on the actual modeled path

How to read: not a simulation — these are the metrics on the one real (modeled) 1970+ history. The genuine high-Sharpe / low-turnover ridge is **~230–270d** (best 250d: Sharpe 0.559, 103 switches); sub-150d windows whipsaw (2-3× the trades) — very costly at 2× leverage and in a taxable account, which is why the pre-tax near-tie at ~160d loses after tax (§9). Green row = recommended 250d.

Columns: **CAGR** return · **Vol** volatility · **Sharpe** risk-adjusted · **MaxDD** worst loss · **Calmar** CAGR/|MaxDD| · **InMkt** % time invested · **Switch** # of in/out trades.

window	CAGR	Vol	Sharpe	MaxDD	Calmar	InMkt	Switch
50	10.7%	21.1%	0.413	-50.1%	0.21	66.4%	381
75	10.0%	21.4%	0.376	-59.1%	0.17	69.2%	309
100	9.7%	21.4%	0.358	-60.4%	0.16	70.3%	269
110	11.8%	21.3%	0.461	-56.8%	0.21	70.5%	233
120	12.7%	21.3%	0.501	-56.9%	0.22	70.9%	215
130	12.3%	21.4%	0.480	-53.6%	0.23	71.2%	205
140	13.1%	21.6%	0.514	-53.3%	0.25	72.2%	185
150	13.4%	21.7%	0.527	-53.7%	0.25	73.1%	175
160	13.8%	21.7%	0.542	-45.4%	0.30	73.7%	161
170	13.7%	21.9%	0.532	-36.0%	0.38	74.5%	147
180	13.4%	22.0%	0.520	-38.9%	0.35	74.6%	147
190	13.3%	22.1%	0.512	-39.9%	0.33	74.9%	145
200	12.9%	22.1%	0.494	-40.3%	0.32	75.2%	145
210	13.7%	22.2%	0.529	-40.3%	0.34	75.6%	127
220	14.0%	22.2%	0.540	-39.5%	0.36	75.8%	125
230	14.4%	22.3%	0.554	-39.8%	0.36	76.1%	115
240	13.9%	22.5%	0.530	-48.7%	0.29	76.2%	111
250	**14.6%**	**22.5%**	**0.559**	**-45.9%**	**0.32**	**76.1%**	**103**
255	14.3%	22.5%	0.545	-47.9%	0.30	76.2%	107
260	14.4%	22.5%	0.551	-47.9%	0.30	76.4%	103
270	14.2%	22.6%	0.541	-48.4%	0.29	76.5%	103
280	13.6%	22.6%	0.513	-49.9%	0.27	76.7%	107
290	12.6%	22.7%	0.466	-55.2%	0.23	76.7%	113
300	12.5%	23.0%	0.457	-59.5%	0.21	76.9%	107
320	12.2%	23.2%	0.439	-67.9%	0.18	77.2%	103
340	12.7%	23.3%	0.460	-62.3%	0.20	77.7%	93
360	12.8%	23.5%	0.460	-61.9%	0.21	78.0%	85
400	12.1%	23.7%	0.427	-51.3%	0.24	78.5%	85

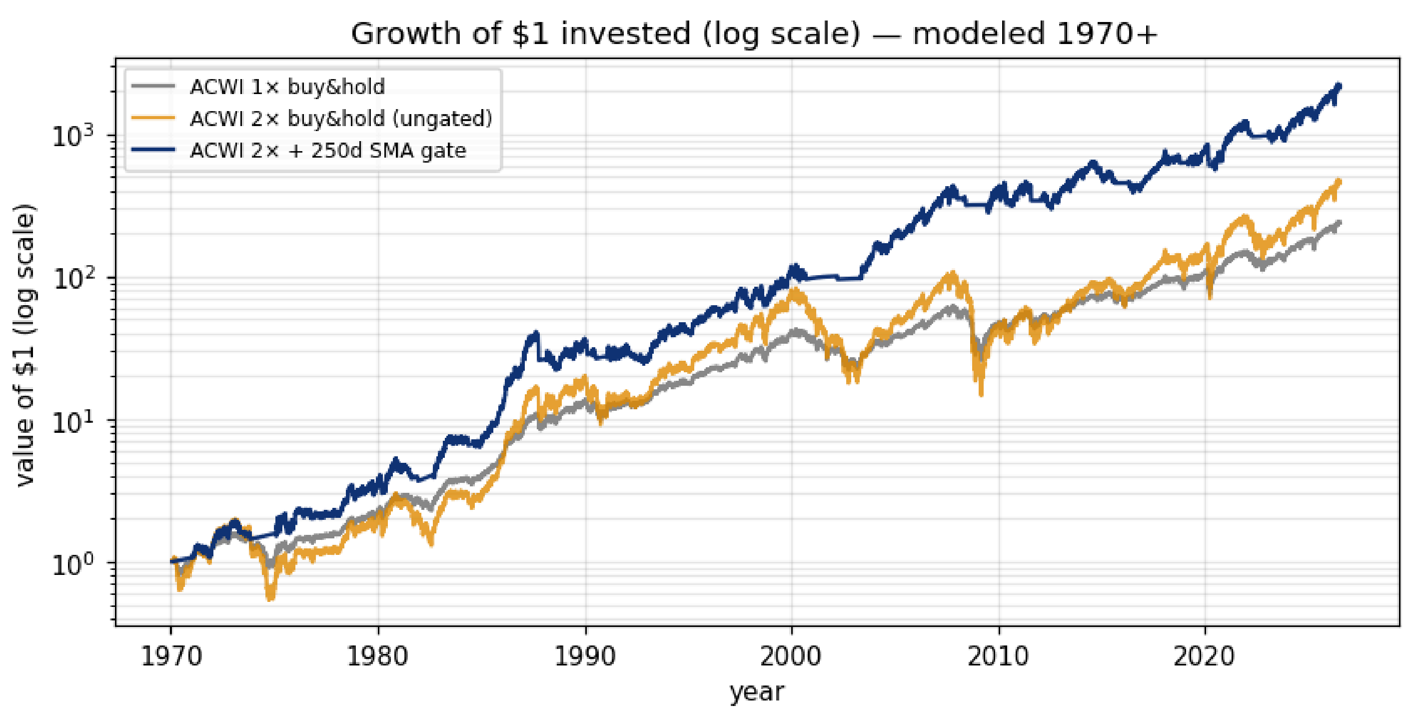
6 · Monte-Carlo of the chosen 250-day strategy



How to read: the spread of outcomes for the 250d gate across the 1500 bootstrapped histories — left = Sharpe (dashed line = ungated 2× for reference), right = max drawdown. The table below gives those distributions as percentiles (p5 = a bad-luck history, median = typical, p95 = a lucky one).

metric	p5 (unlucky)	p25	median	p75	p95 (lucky)
Sharpe	0.23	0.37	0.46	0.57	0.72
CAGR	7.2%	10.3%	12.5%	15.1%	18.9%
Max DD	-70.2%	-59.2%	-53.0%	-47.4%	-41.9%

Beats **ungated 2× ACWI** Sharpe (0.294) in **88%** of paths. (Bootstrap medians read below the actual-path numbers because resampling degrades trend structure — a conservative lens for a trend strategy.)



How to read: growth of $1, log scale (so a straight line = constant % growth). Grey = unleveraged ACWI; orange = raw 2× (note the violent crashes); blue = 2× behind the 250d gate — similar end-wealth to raw 2× but far smoother.

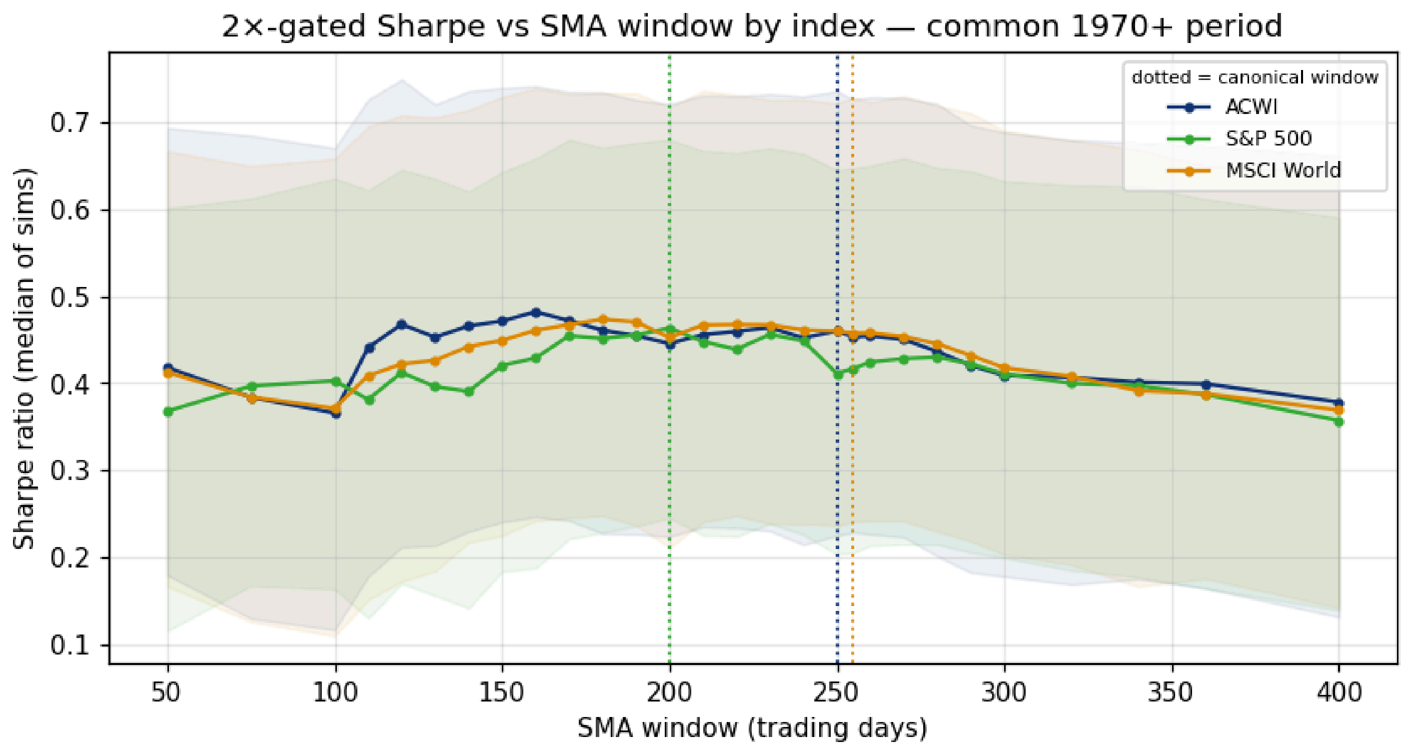
7 · Strategy comparison (actual modeled path, 1970-02-02→2026-07-07)

strategy	CAGR	Vol	Sharpe	MaxDD	Calmar
ACWI 1× buy&hold	10.2%	16.2%	0.507	-58.5%	0.17
ACWI 2× buy&hold (ungated)	11.5%	32.4%	0.294	-86.5%	0.13
ACWI 2× + 250d SMA (recommended)	14.6%	22.5%	0.559	-45.9%	0.32
ACWI 2× + 250d (real-product drag, XS2D-calibrated)	13.9%	22.5%	0.530	-46.1%	0.30
S&P 500 2× + 200d SMA	14.9%	23.4%	0.550	-39.9%	0.37
MSCI World 2× + 255d SMA	14.5%	22.7%	0.550	-54.2%	0.27

The first three rows are ACWI (1× buy&hold, 2× ungated, and the recommended 2×+250d gate). The last two are the **canonical US and World strategies** for reference — 2× **S&P 500** gated by its own 200-day SMA, and 2× **MSCI World (URTH)** gated by its own 255-day SMA — each on the same modeled 1970+ window. Actual modeled path, not a simulation. (§8 runs the same three indices through a matched-path Monte Carlo with beat-probabilities.)

8 · Head-to-head: 2× ACWI vs 2× S&P 500 (200d) vs 2× MSCI World (255d)

The same gated mechanic and the **same 2× leverage** applied to each index at its canonical window, on the **common 1970+ period** — so any difference reflects the *index*, not the leverage or the dates. (Caveat: pre-1988 our ACWI series ≡ World by construction, so the ACWI-vs-World comparison is driven by 1988+ — before that it measures only the 250d-vs-255d window difference.) (Note: your live S&P sleeve is actually 3× SPXL and World is 2× WLDU; here all three are 2× for a like-for-like read.)

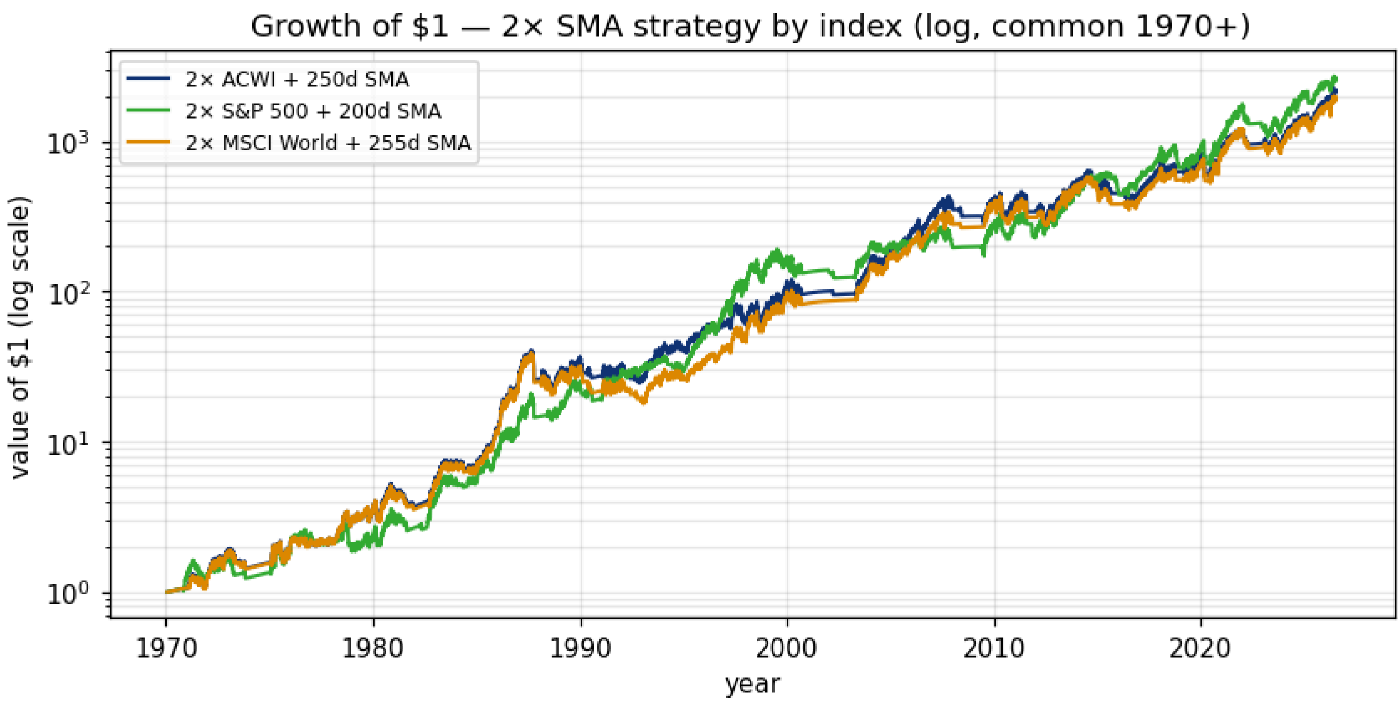


How to read: median 2×-gated Sharpe vs window for each index, with the 5–95th-pct band; dotted verticals mark each canonical window. The peaks confirm the rule of thumb — S&P is fastest (~200d), World/ACWI slower (~255–260d), because adding ex-US and EM makes the trend smoother and rewards a longer lookback.

index (2×, gated)	window	CAGR	Sharpe	MaxDD	Calmar	MC Sharpe median [p5,p95]	P(ACWI beats)
ACWI	250d	14.6%	0.559	-45.9%	0.32	0.460 [0.24, 0.73]	—
S&P 500	200d	14.9%	0.550	-39.9%	0.37	0.459 [0.24, 0.70]	51%
MSCI World	255d	14.5%	0.550	-54.2%	0.27	0.456 [0.23, 0.73]	56%

First four metric columns = actual modeled path; **MC Sharpe** = matched-path bootstrap (same 1500 resampled histories fed to all three indices); **P(ACWI beats)** = share of those matched histories in which 2× ACWI's Sharpe exceeds that index's.

Across matched bootstrap histories, 2× ACWI's Sharpe exceeds 2× S&P 500 in **51%**, 2× MSCI World in **56%**. The three are close — ACWI's edge is diversification (lower single-country concentration), not a higher headline Sharpe.



How to read: growth of $1 for each index's 2×-gated strategy on the common period (log scale). All three are strong; ACWI sits between the US and World lines, which is exactly what a global blend should do.

9 · After costs & taxes (the honest numbers)

Everything above is **frictionless and pre-tax** — an upper bound. This is a German taxable account, so each in→out switch is a realisation event (Abgeltungsteuer 26.375%, ~18.5% effective after the 30% equity Teilfreistellung), while ungated buy&hold defers tax indefinitely (only the annual Vorabpauschale applies). Below: 10 bp round-trip transaction cost per switch + the full German tax overlay (loss carry-forward, Vorabpauschale; via research/tax_overlay.py).

strategy	pre-tax CAGR	pre-tax Sharpe	post-cost CAGR	net CAGR (cost+tax)	net Sharpe	switches
ACWI 1× buy&hold	10.2%	0.507	10.2%	10.0%	0.492	1
ACWI 2× ungated	11.5%	0.294	11.5%	11.3%	0.287	1
ACWI 2× + 250d	14.6%	0.559	14.4%	11.4%	0.409	104
ACWI 2× + 250d (real-product drag)	13.9%	0.530	13.7%	10.8%	0.385	104
ACWI 2× + 200d	12.9%	0.494	12.6%	9.9%	0.352	146
ACWI 2× + 255d	14.3%	0.545	14.1%	11.1%	0.394	108

The gate's turnover is its tax weakness: ungated 2× defers tax (compounds pre-tax, paying only the small annual Vorabpauschale), while the gated sleeve realises — and is taxed on — gains every time it steps out. **Net of cost+tax the 250d gate's CAGR is 11.4% — still ahead of ungated 2×**

(11.3%), though the turnover tax shrinks the pre-tax edge. A faster gate (200d, 146 switches) gives even more back to the tax man. The takeaway: at 2× in a German taxable account the gate is bought for its **drawdown/Sharpe**, not for extra compounding. **Switch-cost sensitivity** (net CAGR of the 250d strategy at 0 / 10 / 25 bp round-trip):

round-trip cost	0 bp	10 bp	25 bp
net CAGR (250d)	11.6%	11.4%	11.1%

Classification risk quantified: if the final fund fails the ≥51% physical-equity quota and gets **0% Teilfreistellung**, the 250d strategy's net CAGR drops from 11.4% to **10.6%** (net Sharpe 0.37) — check the Anlagebedingungen before buying.

10 · Limitations & robustness (read before acting)

This study was hardened by an adversarial review; the material caveats:

- **Window choice is robust, the exact number is not.** Paired matched-path bootstrap of the Sharpe *difference* (250d minus each):

comparison	mean ΔSharpe	SE	t-stat
250d vs 160d	-0.018	0.002	-11.0
250d vs 200d	+0.016	0.001	13.5
250d vs 230d	-0.004	0.001	-4.6
250d vs 270d	+0.010	0.001	15.3
250d vs 300d	+0.047	0.001	47.4

  Read the magnitudes, not the t-stats. Because all windows see the same resampled paths, matched sims shrink the standard error to ~0.001, so even a trivial Sharpe gap shows a large t — that flags *statistical*, not *economic*, significance. Economically the differences within the 250±20d neighbourhood are negligible (|ΔSharpe| ≤ 0.010). Faster windows are a turnover story: on the real path the 200d gate gives up ΔSharpe 0.016 pre-tax and trades 145× vs 103× for 250d; the bootstrap argmax (160d) is likewise a pre-tax, pre-cost artifact of block-resampling — §9 shows the longer window keeps more after German tax. **Bottom line: pick ~230–270d; avoid ≤200d.**
- **Block-length sensitivity.** MC argmax window at bootstrap mean-block 252 / 504 / 756 days:

mean block	252d	504d	756d
argmax window	160d	160d	160d

  The selection is anchored to the *deterministic* sweep (which preserves real trend structure); the bootstrap is a robustness check only, since block-resampling necessarily degrades the multi-month autocorrelation a long SMA exploits.

- **Real-data-only cross-check.** Running the identical 2× sweep on *only* the real ACWI 2001+ segment (no modeling at all) gives best window **230d** — consistent with the full modeled result, so the pre-2001 model isn't driving the conclusion.
- **Leverage scope.** Everything is fixed at 2×, and no 2× ACWI ETF currently exists (the sleeve is synthetic). Best window by leverage:

leverage	best window	Sharpe
1.5×	250d	0.643
2×	250d	0.559
2.5×	250d	0.497
3×	250d	0.445

  The long-window preference holds across leverage; "~250d" is contingent on ~2×.
- **1970–1987 is MSCI World as-is — drift *and* texture.** ACWI only exists from Dec-1987; the head segment (~30% of the trimmed sample) is its predecessor index, unadjusted. ACWI-specific evidence starts 1988 (and daily EM texture only 2003+).
- **1988–2001 daily *texture* is MSCI World's, not ACWI's.** Monthly drift is the real EM-weighted ACWI (validated), but intra-month daily wiggles in 1988-2001 borrow World's shape (no daily EM exists pre-2003). Shown immaterial for a slow gate (±0.02 Sharpe test on 2003+), and the real-only cross-check above confirms it.
- **Financing is a modern best case.** The LETF borrow (3m T-bill + 40 bp × 1.1 swap) reflects a cheap modern swap fund; pre-2008 and a thin all-world 2× would cost more — add ~50–100 bp to financing for a conservative read.
- **Fixed 2% risk-free in Sharpe.** Realised T-bills averaged ~3% over the period, so absolute Sharpes here run ~0.05 high vs an external quote — fine for *ranking* windows (the bias is common to all), not for comparing to other sources.
- **No look-ahead.** The SMA uses the same-day close to decide, but the position is executed the next day (state shifted by one); verified in code.

11 · The real product & portfolio fit

The instrument. The **Scalable MSCI AC World Leveraged Daily Swap Xtrackers UCITS ETF** (LEI 254900NW0MAOB3TRMY48, Luxembourg, registered 2026-05-29) is the first investable 2× MSCI ACWI vehicle — a Scalable Capital × DWS/Xtrackers follow-up to their unleveraged SCWX (€600m+). Filed, not yet trading; final TER/ISIN pending. Its structural template, the Xtrackers S&P 500 2x Leveraged Daily Swap (2010+): TER 0.60%, unfunded swap, USD fund currency with EUR listing — and, decisive for a German taxable account, **classified as an equity fund with the 30% Teilfreistellung**. Our §9 tax numbers assume exactly that (verify the Aktienfonds classification in the final Anlagebedingungen/prospectus before buying). By contrast, the US-listed alternative WLDU (Themes/Leverage Shares 2× Long World Stock Daily ETF — a 1940-Act **swap-based ETF on Vanguard Total World**, so it does include EM; TER 0.75%) holds swaps plus collateral rather than ≥51% physical equities, so it should expect **0% Teilfreistellung** in Germany — the UCITS wrapper is worth roughly TFS × tax rate ≈ 8 pp of every realised gain. §9's bottom row quantifies the same risk for the new fund itself, should its final structure miss the Aktienfonds quota (verify the ≥51% Kapitalbeteiligungsquote commitment in the fund's Anlagebedingungen/prospectus — not the KID, which doesn't state it — before buying).

Execution reality. A UCITS ETF is **not tradeable on Alpaca**, so the bot cannot automate this sleeve. Realistic setups: (a) hold it manually at a German broker (e.g. Scalable) and let the bot's daily job only *signal* the 250d gate via Telegram; (b) automate an approximation on Alpaca with WLDU (2× World ETP — but worse German tax, no EM); or (c) wait for EU broker-API support. The strategy's ~2 switches per year make manual execution genuinely practical.

Correlation with reference strategies (daily returns of the 250d-gated 2× ACWI vs proxies, one common window 1989-08-03 → 2026-07-07 — the start is set by KMLM's 1988 history + warmup):

reference strategy (proxy)	corr
World 2× + 255d (WLDU-gate style)	+0.90
SPXL SMA sleeve (3× SPY, 200d gate)	+0.77
ACWI 1× buy&hold	+0.65
F4 proxy (40% 2×World / 30% gold / 30% TLT)	+0.60
HFEA proxy (45/25/30 3×SPY/3×TLT/KMLM)	+0.53

HFEA/F4 blends are daily-rebalanced approximations of the real quarterly-rebalanced sleeves (GOLY proxied by gold). Dual Momentum, Regime SSO, AAA and 9-Sig are too path-dependent for a quick proxy — the formal correlation matrix runs in mega_backtest.py at promotion time.

High correlation to the US/World trend sleeves is expected — this is the same trade on a broader index. The honest framing: this strategy is a **substitute/upgrade for world-equity trend exposure** (more diversified index, UCITS tax wrapper), not a new diversifier. A formal promotion decision (correlation matrix vs all 7 sleeves, portfolio what-if injection, Monte-Carlo vs deployed mix) should run through mega_backtest.py once the fund has an ISIN and a live price series.